INFOSYS SPRINGBOARD

Receipt & Invoice Digitizer

AI-powered receipt analysis, OCR extraction & expense management

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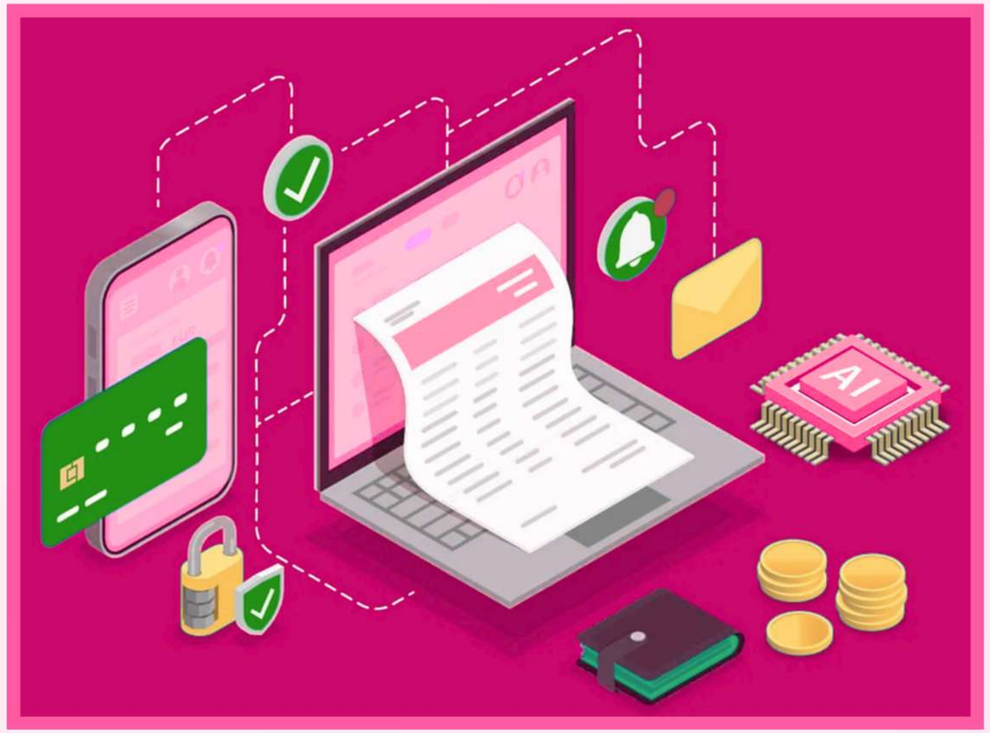
Introduction

Understanding Receipt Digitization

- Receipt and Invoice Digitizer is an AI-powered web application that automatically extracts and stores key data from uploaded receipts and invoices.
- Using OCR technology and a local LLM, it eliminates manual data entry and gives users instant insight into their spending.



Agenda

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Problem Statement

What problem are we solving?

Manual Receipt Management

Businesses and individuals manually sort, store and track paper receipts — a slow, error-prone process.

No Searchable Records

Physical receipts can't be searched or filtered by vendor, date or amount, making tax filing and audits difficult.

Poor Expense Visibility

Without a central dashboard, tracking monthly spending patterns, top vendors and tax totals is impossible.

No AI Assistance

Users have no intelligent assistant to query their own spending data in plain natural language.

Key Features

Receipt Upload

Drag-drop or click to upload images and PDFs with real-time progress indicator

AI OCR Extraction

PaddleOCR + Mistral LLM automatically extracts vendor, date, invoice # and amounts

Analytics Dashboard

Interactive Chart.js graphs showing spending over time with 4 summary stat pills

AI Chatbot

Mistral-powered chat widget answers natural language expense queries in real time

Dark Mode

Full dark/light theme toggle with CSS variables, persisted across sessions

Multilingual Support

i18next integration supporting English, Hindi, Telugu, Tamil, Kannada & Malayalam

Secure Auth

bcrypt password hashing, Google OAuth, session management, forgot/reset password

Admin Dashboard

Admin view of all users, upload stats, activity overview and per-user receipt history

Tech Stack Overview

Flask
Python Web Framework

Python 3 / Flask
Backend Framework

PaddleOCR
OCR Engine

Ollama + Mistral
Local LLM / AI Chatbot

SQLite
Database

Google OAuth 2.0
Authentication

i18next
Multi-language i18n

HTML5 / CSS3 / Vanilla JS
Frontend

Implemented Modules

User Auth Module

Registration, Login, Forgot/Reset Password, Google OAuth 2.0 sign-in. Passwords hashed with bcrypt.

Bill Digitizer Module

Upload receipt images or PDFs. PaddleOCR extracts text; Mistral LLM parses vendor, amount, date, tax.

Expense Dashboard

View all digitized receipts. Filter, search, and manage bill history with spending analytics.

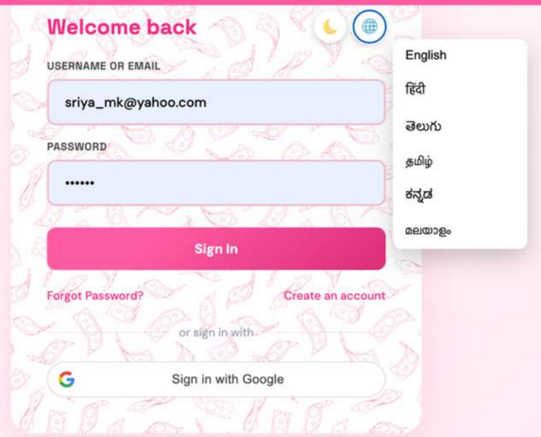
Admin Module

Admin dashboard to view all users and their uploaded bills. Manage user data and receipts.

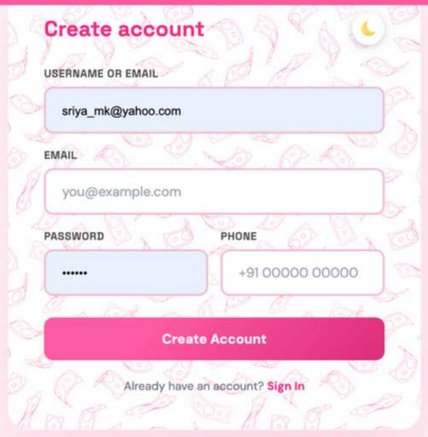
AI-Powered Receipt Processing & Analytics Platform

Key Outputs

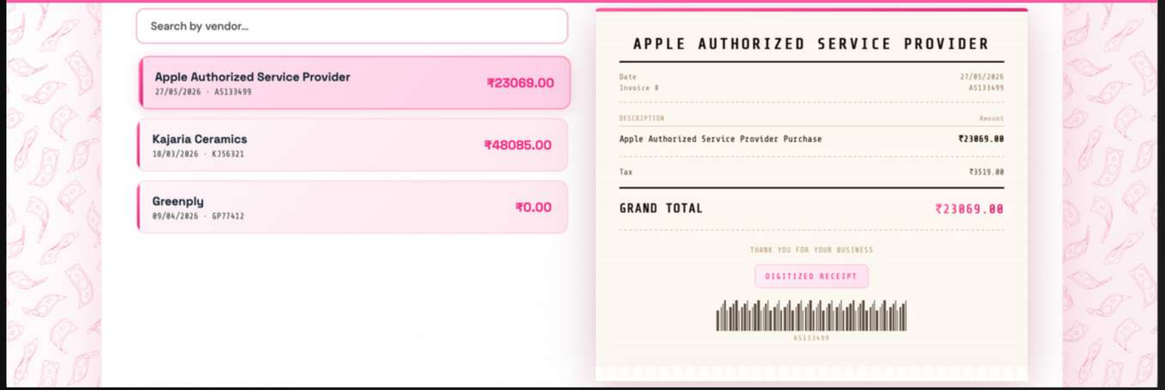
Login Page



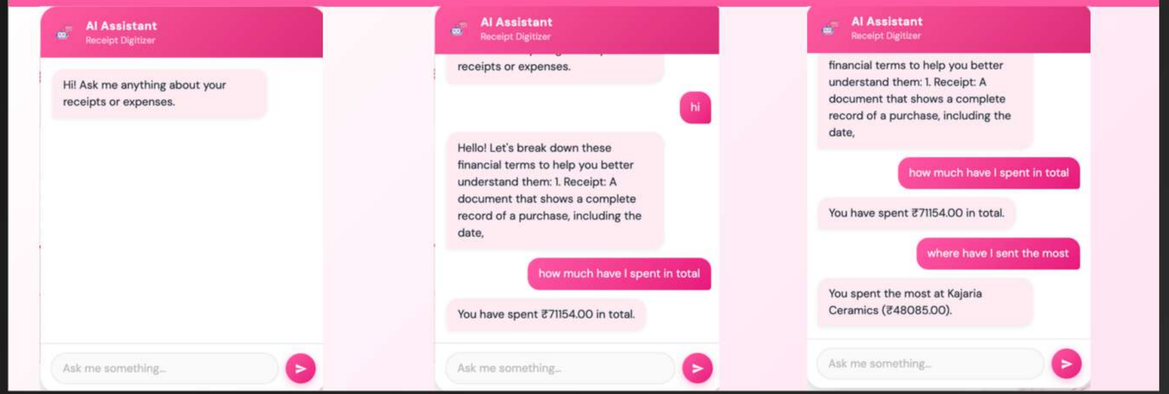
Registration Page



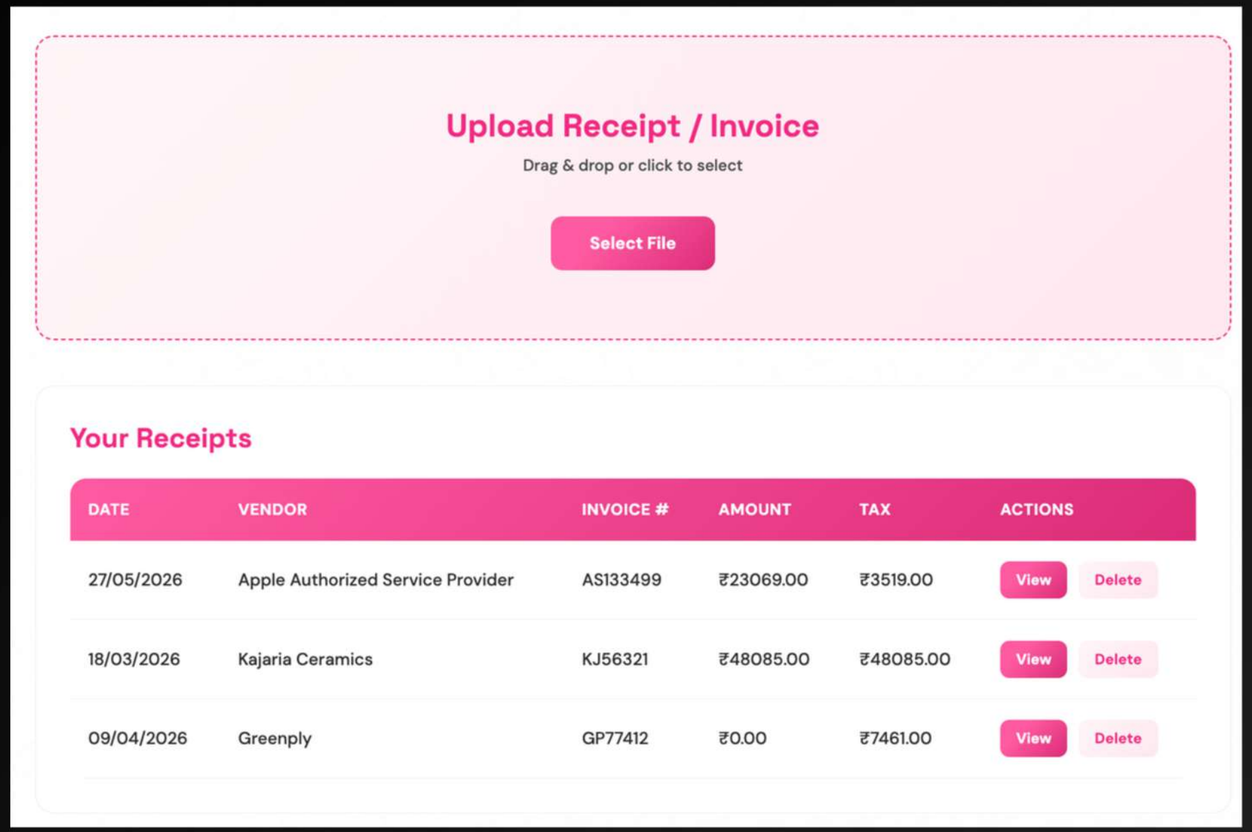
Bill Digitizer



AI Chatbot



User Dashboard



Admin Dashboard

24
Total Users

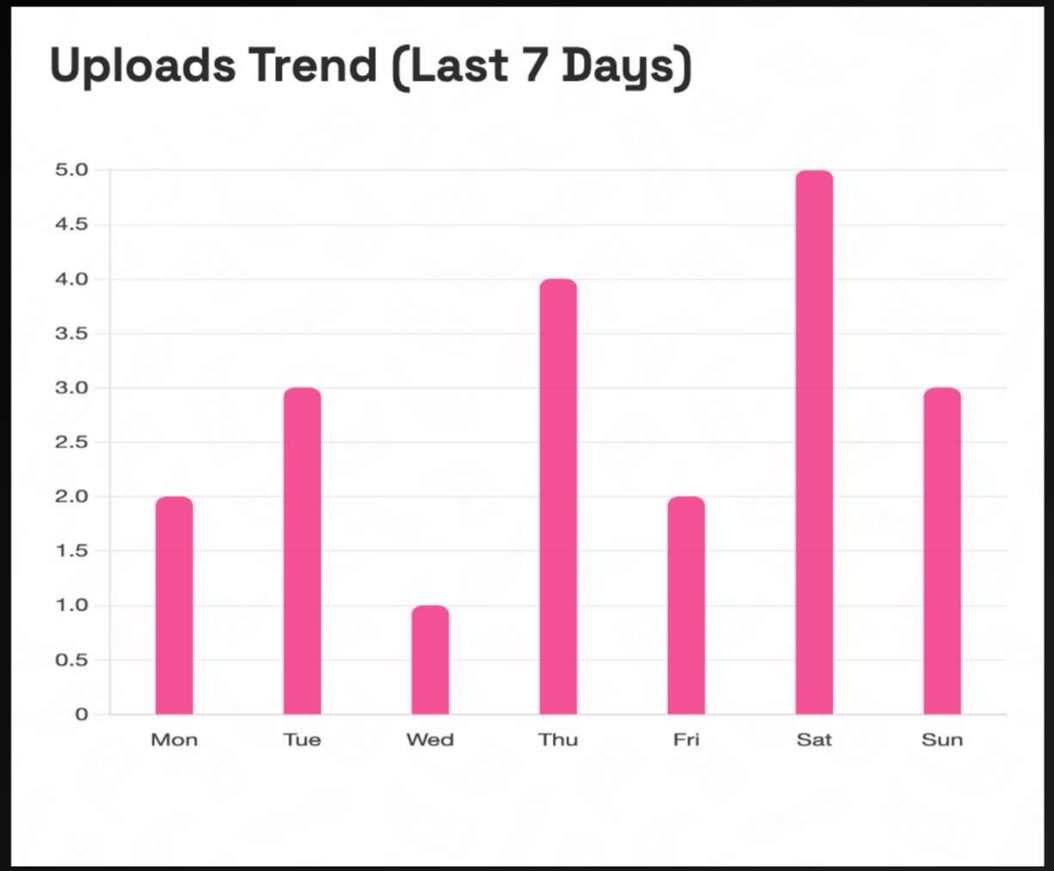
187
Total Bills

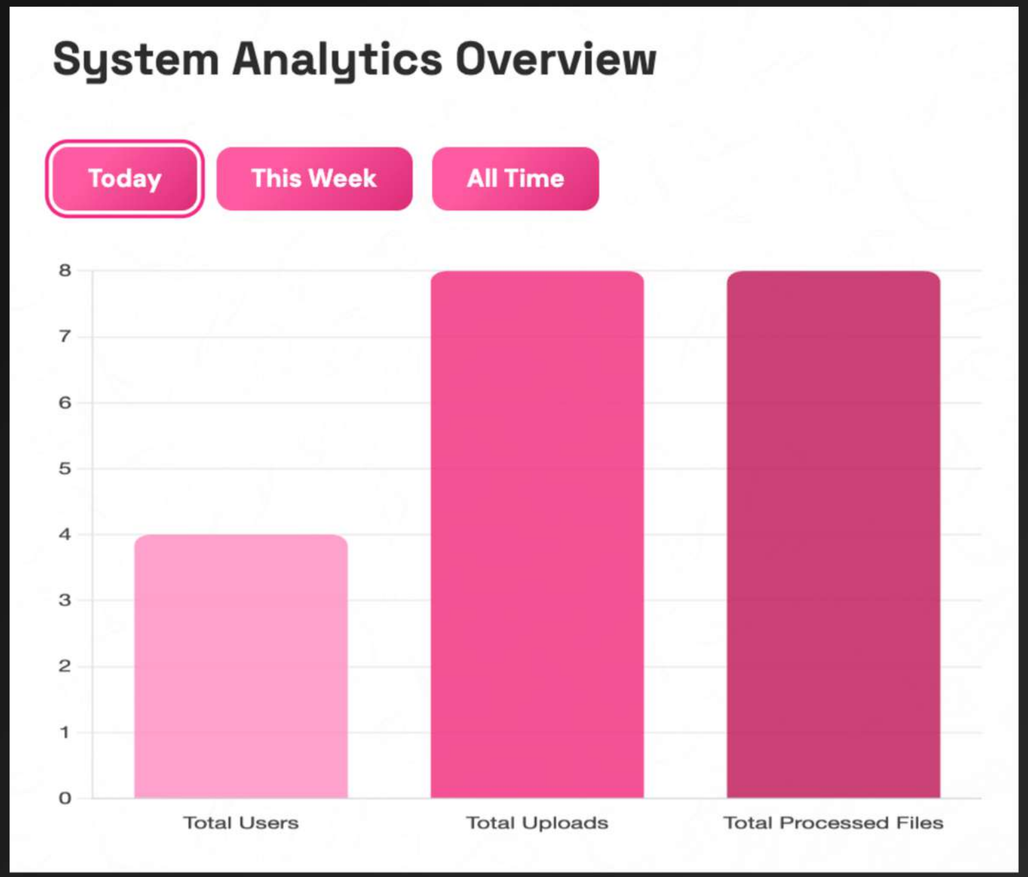
8
Total Processed Files

Username	Email	Bills	View
sriya_mk	sriya@gmail.com	32	View Bills
shaik_m	mastan@gmail.com	28	View Bills
trishala_k	trish@gmail.com	19	View Bills
atul_c	atul@gmail.com	15	View Bills

User	Vendor	Invoice #	Amount	Date
Sriya	Apple Authorized Service Provider	AS133499	₹ 23069.0	2026-03-09
Sriya	Kajaria Ceramics	KJ56321	₹ 48085.0	2026-03-09
Atul	Greenply	GP77412	₹ 134.0	2026-03-09
Atul	Kup Coffee Roasters	25570	₹ 1050.0	2026-03-05

Admin Dashboard- Analytics





Key Outputs — OCR & AI Chat

OCR Extraction Result

KAJARIA CERAMICS

Date	18/03/2026
Invoice #	KJ56321

DESCRIPTION	Amount
Kajaria Ceramics Purchase	₹48085.00
Tax	₹48085.00
GRAND TOTAL	₹48085.00

THANK YOU FOR YOUR BUSINESS

DIGITIZED RECEIPT

KJ56321

AI Assistant Chat

financial terms to help you better understand them: 1. Receipt: A document that shows a complete record of a purchase, including the date,

how much have I spent in total

You have spent ₹71154.00 in total.

where have I sent the most

You spent the most at Kajaria Ceramics (₹48085.00).

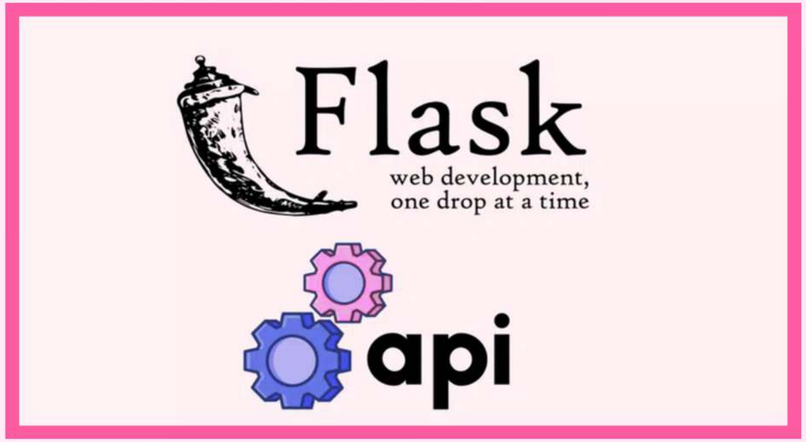
Scalability & Flexibility

Multi-language Support

- The app supports 6 languages — English, Hindi, Telugu, Tamil, Kannada and Malayalam — using i18next client-side internationalization.

System Scalability

- Modular Flask architecture allows easy swap of OCR engine, LLM provider, or database backend.
- The system can scale from local SQLite to cloud PostgreSQL with minimal code changes.



Future Enhancements Overview

Future Enhancements

- Mobile App — React Native app for on-the-go receipt scanning and expense tracking.
- Budget Alerts — Set monthly budget limits; get notified when nearing or exceeding them.
- Cloud Storage — Move from local SQLite to cloud DB (PostgreSQL) with file storage on S3.
- Advanced Analytics — Spending trends, category breakdowns, and month-over-month charts.

Conclusion

- The Receipt & Invoice Digitizer eliminates manual data entry by combining AI-powered OCR with a local LLM to automatically extract structured data from any receipt or invoice.
- It provides users with a secure, multi-language platform to store, search, and analyze their spending, while the built-in AI chatbot makes querying expense data as easy as asking a question.

Thank you!